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18

The Speech Production of Bilingual Adults

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18.1 Introduction

Individuals that learn an additional language often find that it is not a trivial task. Second language acquisition (SLA) research typically refers to the added language as a second language (L2) or a third language (L3), and so forth.[1] An L2 can be acquired in many ways. For instance, one can study a language formally (i.e., in an L2 classroom), informally (i.e., in a naturalistic context), or both formally and informally in an immersion-type context (Saville-Troike, 2005). Among the many difficulties the L2 learner faces, production of the TL is one of, if not the most, difficult, particularly when it comes to the pronunciation of TL sounds (Flege, 2003; Flege & Hillenbrand, 1984; Flege & Wayland, 2019; Oyama, 1976, among many others). This seems to be especially true if the individual in question begins the endeavor of learning the L2 as an adult. In the present chapter, I focus precisely on these individuals, that is, language learners that have become, or are becoming, bilingual as adults. Concretely, this chapter examines the range of factors known to impact speech production in adult learners, such as age of learning (AOL, Flege Munro, & MacKay, 1995; Flege, Bohn, & Jang, 1997a; Flege & Eefting, 1987), length of residency (LOR, Flege et al., 1997a; Flege, 2012; Flege & Liu, 2001), age of acquisition (AOA, Flege & Eefting,

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[1] In the present chapter the term target language (TL) is used to refer to the acquisition of an additional language when it is not particularly relevant to specify what number it is for the learner.

1987), L1/L2 use (Flege, MacKay, & Meador, 1999; Piske et al., 2002), and TL input (Bosch & Ramon-Casas, 2011; Flege, 2012).

More than a half a century of research on language learning has taught us a simple fact: earlier is better. Adults, which I arbitrarily operationalize in this chapter as individuals aged 18 years and older, have a more difficult time learning an additional language when compared with children and adolescents. A central theme in this chapter is that these so-called *age effects* permeate all areas of language learning, but may be more noticeable in the production of the TL. In most cases, native speakers can instantly identify a nonnative accent. Adult learners often find that native-like TL production is extremely difficult to obtain – regardless of whether or not it is a specific goal of the learner – even after years of extensive study and use of the language. Currently, there are many theoretical models used to posit hypotheses that can account for the types of difficulties encountered by language learners, such as the Perceptual Assimilation Model-L2 (Best & Tyler, 2007) or the Second Language Linguistic Perception Model (L2LP, Van Leussen & Escudero, 2015), to name just a few. In the present work, I will not describe in detail these models, but the interested reader can refer to the relevant chapters of this volume (Chapters 7–11), as well as Flege and Bohn (2021) for the Speech Learning Model – revised.

The goal of this chapter is to provide an overview of the speech production of bilingual adults. It considers what we have learned from the past half-century plus of empirical work, and, importantly, where research on bilingual speech production in adults is heading in the coming years. While this overview cannot be exhaustive, it will cover many of the principle variables researchers have used to contribute to our understanding of adult bilingual speech production. The remainder of the chapter is organized as follows. First, I provide a brief historical overview of the motivations behind current speech production research. Second, I focus on age effects and ultimate attainment in adults. Next, I discuss seminal research looking at the role of TL input and use in production outcomes. Afterwards, I highlight exciting avenues of research, such as phonetic drift and semantic processing, that are likely to shape future studies. Finally, I conclude by describing current methodological paradigm shifts that I believe will guide future speech production researchers.

18.2 Even Cavemen Had Accents

Accents are not new. For as long as we have been speaking, we have been doing so through the filter of our native phonology. When we attempt to speak another language, our native phonological structure, both segmental and suprasegmental, can predict TL production difficulties (Flege, 1995). In his book *Languages in Contact: Findings and Problems*, Weinreich (1953) considered L1 Russian L2 learners of English and their production of English /t/. In Russian, /t/ is produced with a short interval of time between it and the following vowel, like [ta]. In English, this interval is longer and accompanied by aspiration ([t^ha]). Weinreich (1953) was interested in whether these learners would acquire a new category for the English segment, that is, whether or not they would produce /t/ with aspiration like English speakers do. He proposed two possibilities: either they would develop a 'coexistant' phonetic system, whereby they would learn to produce both [t] and [t^h] as realizations of a single /t/ category, or they would develop a new category altogether such that [t] was available in Russian and [t^h] in English. Weinreich (1953) considered the possibility that the adult learners would develop a new category for English to be more difficult and less likely.

Throughout this chapter I refer to the term phonetic category, as I did in the previous paragraph, thus, before continuing, it is worth a moment of our time to consider what this entails. In the speech literature, researchers use this term to refer to a vowel- or consonant-like unit (e.g., Flege & Eefting, 1988), though the specifics can vary (see Trubetzkoy, 1969). Importantly, this concept differs from the notion of a phoneme, a mental representation that indicates the minimal unit that can distinguish words, in at least the following ways. First, a phonetic category is language-specific. For instance, two languages can have the phoneme /t/, but have different phonetic categories for the corresponding acoustic realization, for example, [tʰ] or [t]. Next, as illustrated in the previous sentence, a phonetic category operates at the level of the segment. That is, like a phoneme, it represents a single unit that is smaller than a syllable. Lastly, a phonetic category is ascribed a certain level of abstractness in that it does not refer to one specific acoustic realization. If you were to say the word 'taco' in English ten times, the realization of the initial [tʰ] would be slightly different in each iteration, but still pertain to the same phonetic category.

Weinreich's reflections about /t/ exemplify the way in which much of the research on adult speech production has been conducted. That is to say, researchers have been interested in understanding and predicting how TL difficulties can be explained by the organization of one's native phonology. The parameter Weinreich was considering is known as voice-onset time (VOT, Lisker & Abramson, 1964). VOT refers to the duration of the time interval between the release of the stop burst and the onset of voicing of the following segment. A large amount of the extant research on adult speech production includes VOT. This is so because cross-linguistically stop contrasts are produced in varying ways based on VOT. In other words, the same phonological contrast in two distinct languages can be realized in acoustically different ways with regard to VOT, like how /t/ is produced as [t] in Russian and [tʰ] in English.[2]

[2] Other acoustic cues differentiate stop identity. For example, in comparison with voiced stops, voiceless stops have longer closure duration (Raphael, 2005), higher f_0 (pitch) at the release (Kingston & Diehl, 1994), and shorter preceding vowels (Chen, 1970).

In a similar vein as Weinreich (1953), Flege and Hillenbrand (1984) proposed that L2 learners struggle with acoustically distinct L2 segments that are related to L1 categories (i.e., L1 [t] and TL [t^h] as pertaining to an L1 /t/) because of how acoustically similar they are. Flege (1987) set out to test this hypothesis by analyzing the production of /t/ in native French and native English speakers who were learning English and French, respectively, as an L2. The authors assumed that speakers from both groups heard the TL segment as realizations of their L1 category, that is, French speakers hear English [t^h] as an exemplar of French /t/ and vice-versa. Their analysis of VOT in word-initial stops found that the learners produced L1-like VOT values in the TL; however, the productions of both learner groups differed from those of monolinguals in both the learners' L1 and the TL. This led Flege (1987) to conclude that the learners had noticed some of the acoustic differences between the TL segment and their L1 categories. In a related example, Flege and Eefting (1987) examined Spanish speakers' production of word initial stops in English. The study found that the Spanish speakers produced English stops with aspiration, but the acoustic analyses showed that VOT values were longer than in Spanish words, but not as long as VOT values produced by monolingual English speakers. Flege and Eefting (1987) concluded that the learners had developed novel L2 phonetic categories for English, though they might have differed from those of the monolingual controls because of exposure to Spanish-accented English.

At this juncture you may be thinking that differences such as these seem trivial, and that may be true for some learners; nonetheless, they have important implications with regard to both speech production and perception (see Chapter 17, this volume), and they provide insight as to how speech research is often conducted with bilinguals. The fact that languages have varying acoustic patterns for similar sound categories – not just with stops, but also with other vowels and consonants – provides virtually infinite possibilities for examining bilingual speech. As we will see, L2 learners are likely to produce (and perceive) TL sounds using the distributions of the acoustic parameters of their L1, rather than those of the TL. For instance, as in the example highlighted above, it has been shown repeatedly that beginning L2 learners will produce TL /ptk/ with VOT based on the values for these segments in their L1 (Caramazza et al., 1973; Flege & Eefting, 1987; Flege & Port, 1981; Port & Mitleb, 1983). More generally, as TL productions deviate from native values, they are more likely to be characterized as "foreign sounding" (González-Bueno, 1997; Sundara, Polka, & Baum, 2006). The good news is that experience with the TL has proven to be correlated with increased accuracy in production (Aoyama et al., 2004; Flege, 2003, among many others), and accurately producing something seemingly minor, like the stop contrasts of the TL, can significantly improve foreign accent ratings (Sundara et al., 2006).

These early studies fostered some of the important questions that shape bilingual speech production research on adults to this day. For example, how and why does age affect speech production outcomes and is it possible to become native-like? What is the role of TL input and use in acquiring the sounds of another language? How do TL sound categories develop over time? How is phonological learning modulated by the context in which one learns? Or how does speech production development relate to that of speech perception? Researchers have attempted to respond to questions such as these for the past half-century. In the next section we will consider age effects and ultimate attainment.

18.3 Is Youth Wasted on the Young?

Many of us have heard, or can attest firsthand, to the fact that children seem to learn additional languages better than adults. We often refer to them as 'sponges', as they seem to effortlessly soak up an L2. Adults, on the other hand, are more likely to display difficulties with seemingly 'easy' aspects of a TL. When it comes to learning the phonology of an L2, there is overwhelming evidence suggesting it is better to start at an earlier age, if the goal is to obtain more native-like production (see Bongaerts et al., 1997; Flege et al., 1999; Flege, 2003; Oyama, 1976, among many others). This notion is hardly controversial, as countless studies show that this task seems to become more difficult after a certain age.

Lenneberg (1967a, 1967b) proposed an innate language faculty as the reason children are able to ‘grow’ language. As his work gained support, the notion of the ‘critical period’ became popular in explaining the ‘age effects’ found in many cognitive abilities, language being one of them. Some researchers still believe that after a certain age – a critical period – it is no longer possible to acquire certain abilities due to constraints associated with neuronal maturation. Nonetheless, with regard to language, many learners manage to defy the odds. In other words, in spite of learning their L2 later in life, many adults manage to acquire TL sound categories. We have all heard hard-to-believe anecdotes about the ‘friend of a friend’ who speaks perfect Italian, indistinguishable from a native speaker, after using Rosetta Stone for a year. But is there any truth to this ‘anecdata’? Work by Flege and colleagues examining an Italian community living in Ottawa for an average of 20 years shows that *some* adult learners can achieve native-like pronunciation despite their late exposure to English (see Flege, Frieda, & Nozawa, 1997b, 1999; Piske et al., 2002). Conversely, there are also cases of early learners with nontarget like speech production. Rather than abandoning the ideas of Lenneberg, findings such as these have led many researchers to develop variations of the original hypothesis (e.g., Scovel, 1969, 1988; Seliger, 1978; Walsh & Diller, 1981). One possibility, for example, is that age effects are related to a sensitive period that varies from person to person much more than initially postulated.

Others ascribe adult language difficulties to the fact that they already begin the acquisition process with a developed phonological system in place. A natural implication of this fact is that the already established system is the source of cross-linguistic influence. For instance, a seminal investigation by Flege et al. (1995) examined the L2 English of Italian immigrants living in Canada. These participants are of particular interest due to the fact that most of them had resided in Canada for several decades (mean = 23 years). Flege et al. (1995) found that age of learning (AOL) was a key factor in explaining their L2 outcomes. Specifically, the earlier the AOL, the more native-like the participants performed regarding production and perception of English. Furthermore, AOL was positively correlated with foreign accent ratings from native English speakers. The results show that most of the adult learners did not have the same success as the early learners, possibly because their native-language categories were already firmly established in long-term memory.

Similarly, Flege et al. (1998) examined the production of /t/ in learners of American English. This study included Spanish speakers who were partitioned into two groups based on their age when they began learning English (under or over 21 years old), and compared them to a group of monolingual English speakers. Flege et al. (1998) utilized multiple regression to account for variation in VOT for /t/ in utterance-initial position, where we would expect [t^h], in 60 words.[3] The model included factors that were lexical or phonetic in nature, along with predictors related to linguistic experience. The lexical factors were based on the participants' ratings of the experimental items. Specifically, each participant rated the 60 items for their familiarity, imageability, relatedness to Spanish words, and the age at which they were acquired. The predictors related to linguistic experience were age of exposure to English, age at which the

[3] Multiple regression is a statistical technique that allows the researcher to include multiple predictor variables, both continuous and categorical in nature, in a single model. This technique provides parameter estimates for each predictor in the model. The estimates are often interpreted as an assessment of the strength of the relationship between a given predictor and the outcome variable while holding constant the effects of the remaining predictors. The interested reader is referred to Berry and Feldman (1985) and Lewis-Beck (1980) for a primer on the technique.

participants began to feel comfortable speaking English, length of residency in the United States, use of Spanish, and age. The phonetic predictors included syllabic structure of the word (monosyllabic, disyllabic), and the height of the following vowel (low, mid, high). The participants' productions ranged from target-like (i.e., [t^h]) to nontarget-like (i.e., [t]) regarding VOT values. The analyses showed that VOT was more target-like in monosyllabic words and before high vowels, but the lexical factors did not account for variation in VOT values. The amount of English input, assessed through age of exposure and length of residency, was a better predictor of target-like productions.

We will return to the role of input shortly, but first we will consider the final state or end point of language learning in relation to the learner's ability to attain native-like proficiency in an L2. A large portion of the L2 research on adult phonological acquisition has focused on this notion, which is referred to as ultimate attainment (see Mitchell, Myles, & Marsden, 2019). The main question posed in this line of research deals with how close a nonnative speaker can come to acquiring the phonology of the TL. That is to say, can an adult learner become indistinguishable from a native speaker? In order to answer this question this line of research has necessarily searched for adult bilinguals that have a large amount of experience and have reached high levels of proficiency in their L2 (e.g., Bongaerts et al., 1997; Face, 2018a, 2018b, 2021; Face & Menke, 2020; Stölten, Abrahamsson, & Hyltenstam, 2015). For example, Stölten et al. (2015) examined ultimate attainment by recruiting native Spanish speakers that were highly proficient in Swedish. The researchers partitioned the participants into two groups, early and late learners, and analyzed their production of Swedish /ptk/. Their analysis found that the learners were indeed indistinguishable from native speakers when using a raw VOT measurement, that is, in milliseconds; however, when taking speech rate into account by calculating VOT as a percentage of word duration, they found that few of the late learners produced /ptk/ in a native-like manner. This study suggests that adult learners can come quite close to native-like production, but when using stringent analytic techniques, production differences emerge, at least with regard to voice-timing.

A series of studies by Face and colleagues explored ultimate attainment of Spanish in American English speakers who had been residing in Spain for up to 60 years (mean = 36). These studies compared the adult learners' production of voiced stops (Face, 2018a; Face & Menke, 2020), rhotics (Face, 2018b), and laterals (Face, 2021) to those of native Spanish speakers. Each of these cases present difficulties for native English speakers due to cross-linguistic articulatory differences. In all four studies, the researchers found that the adult learners' production differed from that of native Spanish speakers, though there were considerable individual differences. For each of the segments being analyzed, there were participants with production values within native ranges, with the exception of VOT in Face and Menke (2020). This begs the question as to what might be different among phonetic variables, such that some appear to be harder to attain for learners than others. The answer to this question is likely complex with multiple factors simultaneously at play, such as, for instance, the status of the segments in question in the learner's native language.

Taken together, these investigations on ultimate attainment show that adult learners generally differ from native speakers, even when they are highly proficient in the TL and have decades of experience. This is particularly true when conducting group comparisons; however, the majority of the aforementioned studies show that some individuals do become native-like with regard to the parameters measured. The results support the hypothesis that speech learning can indeed continue throughout the lifespan, though it is not entirely clear to what this learning is attributed. This research suggests that other factors, aside from age of learning, are likely at play.

18.4 The Roles of Input and Use

In much of the research that has been conducted on adult learners, input has not been considered the most important factor. Perhaps difficulties producing a TL accurately can be attributed in part to the quality and/or quantity of the input learners receive. Learners who acquire their L2 in a naturalistic setting, for instance, are likely exposed to input that differs with regard to individuals who learn in a formal classroom setting. As we have seen thus far, the focus in SLA research has been on cross-linguistic interactions between sound systems, individual differences, maturational states, as well as AOA and LOR. This fact seems to be at odds with the relative importance that input receives in L1 acquisition theory and research. At this time, the role input plays in the acquisition of L2 phonetic categories in adult learners remains unclear. Another understudied factor is the amount of L1 and TL use. It seems reasonable to assume that using the TL is a critical element for progressing in all areas of language learning, but how much and how often does the TL need to be used for optimal outcomes, and to what extent does L1 use play a role? Could it be a detriment to L2 progress? Perhaps L1/L2 use account for the improvement seen in the aforementioned studies by Flege and colleagues examining Italian immigrants in Canada?

In a follow-up to the Flege et al. (1995) foreign accent study, Flege and MacKay revisited the adult learners from the original investigation (as cited in Flege, 2012). One-hundred and sixty of the 240 participants were rerecorded to see if these highly proficient adult learners had improved in their English production over the next ten years. The average LOR at the second time of testing was forty-three years. Three sub-groups were established based on changes in self-reported use of English at the two times of testing (in 1992 and in 2003). The first group reported using English 28% less at the second time of testing, the second group reported no change in English use, and the third group reported using English 24% more at the second time of testing. The researchers found that increased use of English was associated with more native-like VOT (determined by an increase from 1992 values) for voiceless stops. A decrease in English usage resulted in no change in VOT production.

In an unpublished longitudinal study on 15 native Spanish late learners who immigrated to the United States, Flege found no significant improvement in the foreign accent ratings of their English pronunciation after a period of five years in Alabama (as cited in Flege, 2012). However, a post-hoc analysis of the data provided several insights with regard to the role of input. Flege compared the three 'worst' (more foreign accented) learners with the three 'best' (least foreign accented) learners and found that the large disparity between groups could be explained by the fact that the 'best' learners were those who reported using English more, specifically in contexts where it could be considered optional (i.e., when having conversations with their friends). Presumably these learners received more input in these situations, though Flege only provided impressionistic inferences. Studies controlling for TL input and use can shed light on the relative importance of these factors as they pertain to speech production in adult learners.

One way researchers have attempted to disentangle the effects of input and use is by analyzing the context in which language acquisition occurs. Despite the relatively long history of study abroad (SA), only in the past twenty-five years have researchers begun to seriously consider the benefits immersion learning has on L2 outcomes (Freed 1995), and how these benefits directly compare to those offered in the traditional, at-home L2 classroom context (AH).[4] There is evidence that L2 production gains are positively correlated with length of stay (Stevens, 2001) and occur at a faster rate than in the AH context for conversational speech (see Díaz-Campos, 2004, 2006). These findings suggest that TL input may well be the key to native-like production; however, it is difficult to know what type of input the participants in the SA context receive and how it compares

[4] For a complete overview, see Chapters 23 and 24 of this volume.

to the input available to at-home learners. For example, it is possible for SA learners to mainly interact with students from their home country, or use the TL minimally (e.g., Gorba & Cebrian, 2021). Alternatively, improvements seen in the SA research could be due to increased usage of the TL, as most learners in the at-home context only use the TL for a few hours each week. Domestic immersion programs represent an understudied context in which L1/L2 use and input can be more easily accounted for (see Casillas, 2020a, 2020b).

18.5 What about those Suprasegmentals?

The studies outlined thus far point to the importance of several key experiential variables – L1 phonology, AOL, TL input, and amount of L1/L2 use – and their effects on the production of L2 segments.[5] Our discussion has not gone beyond the segment. That is to say, we have not yet considered L2 production of suprasegmental aspects of speech, such as syllables, stress, length, tone or intonation. This is indicative of the fact that the primary focus of research on bilingual speech production in adults over the past half-century has been at the segmental level. As evidence of this, consider the fact that many of the major theoretical models describing L2 phonological acquisition do not have specific hypotheses regarding how suprasegmentals are acquired. That being said, some researchers have used the current models and extended them to make predictions about the acquisition of suprasegmental aspects of speech. For example, Trofimovich and Baker (2006, 2007) analyzed how L2 language experience modulated the production of prosody and fluency in adult L1 Korean learners of English. In line with the general findings we have seen at the segmental level, learners with more L2 experience produced suprasegmental aspects of English

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[5] Note that some researchers distinguish between AOL and AOA. For some, the former is reserved specifically for adult L2 learners. This distinction represents a terminological difference that reflects the view that one ought to distinguish between formal, conscious learning and informal, unconscious acquisition, though many researchers use the terms interchangeably. For discussion on the topic, see Mitchell et al. (2019).

in a more native-like manner when compared with learners with less L2 experience. Furthermore, the authors affirmed that suprasegmentals can contribute to foreign accent in the same manner as segmentals. They conclude that models of L2 speech learning (e.g., Chapters 7–11 of this volume) can be extended to account for L2 suprasegmental production (see Mennen, 2015, for example). Our discussion of suprasegmental aspects of adult speech production is limited in scope. For a complete overview, see Chapter 21 of this volume.

18.6 Into the Unknown

Now I will briefly highlight some of the more recent lines of research relating to speech production in adults that are likely to inspire future studies. While the majority of the investigations we have considered have focused on the influence one's native language has on the production of an L2, some studies have examined how acquiring an L2 can affect the production of one's native language. This bi-directional influence bilingualism has on the L1 is referred to as phonetic drift and has been documented in individuals with ample L2 experience (see Flege, 1987, 1995, 2002, among others) as well as in early bilinguals (e.g., Mora & Nadeu, 2012) and switch dominance early learners (e.g., Casillas, 2015; Casillas & Simonet, 2016). What is relatively novel, however, is the finding that phonetic drift can also occur during the beginning stages of adult sequential language learning. Chang (2012, 2013) found that native English speakers learning Korean as an L2 assimilated phonetic properties of Korean into their native English after initial exposure in an immersion program. The early phonetic drift effect has been replicated with some degree of success. For example, native English-speaking learners of Spanish have displayed phonetic drift after six weeks of exposure in the AH context (Herd et al., 2015; Huffman & Schuhmann, 2015a, 2015b). Lang and Davidson (2019) did not find the effect in English-speaking learners of French after the same period of time in a SA program, however, longer-term learners from the same study did show indications of L2 influence on their English vowel space. Future research in this area with novel language pairs can shed light on how TL input and use modulate phonetic drift in adult speech production. For a complete overview of the topic, the interested reader is referred to Chapter 33 of this volume.

Other lines of research on L2 production have considered the nature of the representations acquired by adult learners. One question posed by this line of research revolves around whether or not the distinct types of cognitive processes utilized during speech production can have an influence on L2 phonetic processing. Phonetic processing is operationalized as the "processes involved in the planning, programming, and execution of articulation" (Gustafson, Engstler, & Goldrick, 2013, p. 506). One of the cognitive processes that has been used to assess phonetic processing is semantic processing (Gustafson et al., 2013; Nozari et al., 2010), which requires that one access the meaning of an object, concept, etc. Van Lancker Sidtis, Cameron, and Sidtis (2012) found that patients suffering from Parkinson's disease had difficulties producing speech when participating in tasks that required semantic processing. In this investigation, Van Lancker Sidtis et al. (2012) utilized picture naming to incite semantic processing, and found that the articulatory impairments of the participants were compounded when compared to tasks that did not require semantic processing (i.e., reading and delayed repetition). Recent studies have shown that semantic processing also affects stop production in adult language learners (Casillas, 2019; Gustafson et al., 2013), though the effect appears to diminish as TL proficiency increases. These findings have methodological implications for speech production research, as the nature and amount of cross-language interactions in adult bilingual speech may be underestimated due to reliance on tasks that do not require semantic processing. Future research should explore semantic processing effects on other speech segments.

A final avenue for future consideration is, in fact, quite an old one: the production-perception link in L2 speech. Investigators have spent decades discussing the nature of the relationship between the two modalities. Importantly, if such a relationship exists, current models posit one ought to observe a clear correlation between the production of categories and the perceptual judgments of these categories within listeners (Diehl, Lotto, & Holt, 2004). This does indeed seem to be corroborated in some research on production and perception in monolinguals and bilinguals (Elvin, Williams, & Escudero, 2016; Evans & Alshangiti, 2018; Flege et al., 1997a, 1999; Flege, 2003), though other studies find weak or no evidence for such a correlation (de Leeuw et al., 2021; Gorba & Cebrian, 2021; Kartushina & Frauenfelder, 2014; Peperkamp & Bouchon, 2011; Rallo Fabra & Romero, 2012). As pointed out by Nagle and Baese-Berk (2021), there are methodological concerns that may explain discrepancies in the extant literature. For instance, the current models positing production-perception links have been essential in establishing a framework describing the nature of the relationship between modalities, but fall short of offering a roadmap for how the links can be investigated. As a consequence, researchers have employed a variety of experimental paradigms and analytic strategies that may not be conducive to explaining how production-perception links develop and evolve during L2 learning. Nagle and Baese-Berk (2021) suggest that future research in this area would benefit from: (1) prioritizing longitudinal designs, (2) accounting for different learning scenarios, and (3) including individual differences. Moving forward, our knowledge of bilingual speech production will only benefit from furthering our understanding of the production-perception link.

18.7 The Methods They Are A-changin'

In this final section, I will discuss what I believe are some essential methodological changes that will shape how we conduct speech production research moving forward. First, I will discuss the importance of longitudinal designs. Second, I will consider a recent push for bilingual baselines in L2 research and how it goes hand-in-hand with instructed second language acquisition (ISLA) in shaping how we design our studies. After, I will focus on methodological shortcomings related to power and sample size, and, finally, I will turn my attention to open science practices and reproducible research.

18.7.1 Longitudinal Research Design

In light of the fact that L1 acquisition is a process that develops over the course of *years* during the early stages of child development, a logical assumption is that L2 acquisition in adults is also a process that requires a substantial amount of time. Unfortunately, longitudinal research is costly, requires careful planning, and involves complex analytic strategies (Nagle, 2021). In spite of these difficulties, longitudinal research is invaluable in providing crucial insight on developmental processes. In speech production research, there is a dearth of longer-term, multiwave studies dealing with pronunciation development (Nagle, 2021), though some recent work has explored how L2 phonetic categories develop over the short-term (Casillas, 2020a, 2020b; Holliday, 2015; Nagle, 2017a; Nagle, 2017b, among others). For example, Nagle (2017a) examined bilabial stop production in 26 native English speaking L2 learners of Spanish studying in an AH, traditional context. He analyzed their stop production over the course of an academic year and found that the learners reduced VOT over time, suggesting they may have developed language specific phonetic categories for [b] and [p]. The analyses also revealed that learning generally followed a quadratic trajectory, implying phonetic development occurred towards the beginning phases of testing and then slowed down over time. Longitudinal studies, such as Nagle (2017a), show that the beginning stages of phonetic category formation can take place in a classroom learning context over the course of just one academic year. As longitudinal designs become more prevalent in L2 speech production research we can expect to learn more about the relative difficulty of learning specific segments, contrasts, and suprasegmental aspects of L2 phonology.

18.7.2 Bilingual Baselines

Traditionally L2 acquisition research has used monolingual speakers as comparison groups for L2 learners in speech production (and perception). A recent trend in L2 phonetic research has pushed for the use of a bilingual baseline as the starting point for measuring adult learner outcomes (e.g., Casillas, 2020b; Sakai, 2018), though there has been a push from the field of SLA to move away from idealizing native speakers for nearly forty years (see Bley-Vroman, 1983; Cook, 1999; Ortega, 2013). According to Levis (2005), a predominant ideology in pronunciation research is that "it is both possible and desirable to achieve nativelike pronunciation in a foreign language" (p. 370), which he refers to as the nativeness principle. The idea motivating the methodological shift to bilingual comparison groups is based on the notion that comparing an adult L2 learner with monolingual native speakers sets unobtainable expectations for the learner because it is not possible for an adult to become a native speaker of another language. Perhaps a more convincing argument can be made by restating this idea in a way that highlights the obvious fact that an adult L2 learner cannot become a monolingual in another language, and likely has no desire to somehow lose their L1. For this reason alone, we can begin to see why it makes little sense to compare learners with a monolingual population that is fundamentally different from the population that represents their goal. In other words, the use of bilingual comparison groups can provide a fairer and more useful assessment of the learners' progress because, in essence, the learners' abilities are compared to those of a target population (bilinguals) that represents their learning goal (bilingualism).

This paradigm shift also changes the narrative around the idealization of monolingual, native speakers. In its place, a bilingual baseline approach gives more agency to bilingual populations and helps to normalize all of the characteristics and inherent value of bilingual speech. As noted by Sakai (2018), a consequence of this methodological shift in speech production research is that investigators and language practitioners reevaluate their goals regarding phonetic learning and move towards scholarship focusing on intelligibility rather than nativeness. For a more in depth overview, the interested reader is referred to Chapters 34 and 35 of this volume.

18.7.3 Stepping up our Game

Further methodological concerns include statistical power and sample size. Most studies in the social sciences test for small effects (Ellis, 2010; cf. Plonsky & Oswald, 2014), include small samples sizes, and, therefore, are underpowered. Brysbaert (2020) highlighted how these issues are rampant in bilingualism research, likening it to having blurred vision, and, in a recent call to arms, affirmed that it is now time for us to step up our game. The severity of having a field plagued with low power should not be overlooked, as an underpowered study is more likely to commit a type II error (false negative), and contributes to a literature with lower positive predictive value. Consequently, this implies that the prevalence of significant findings in our field may be indicative of publication bias. Future empirical studies on bilingual speech production will require careful planning and sample size justification if we hope to see clearly.

18.7.4 The Way Forward

In many fields, analytic flexibility in data analysis – commonly referred to as researcher degrees of freedom (Gelman & Loken, 2014; Simmons, Nelson, & Simonsohn, 2011) – can lead to substantially different conclusions based on the same data set, or what is now being referred to as the *inference crisis* (see Rotello, Heit, & Dubé, 2015; Starns et al., 2019). The "garden of forking paths” associated with researcher degrees of freedom directly affects our ability to replicate findings and thus accumulate knowledge. The early 2010s saw the reproducibility crisis take hold of the psychological sciences. A push for open science, reproducible methodology, and increased value of replication studies followed as a consequence. This methodological framework and associated techniques have reshaped research methods in psychology and have slowly but surely made their way into adjacent fields. This is certainly the case for applied linguistics, second language acquisition, and phonetics and phonology. More and more, journals in these areas are *requiring* open science practices, such as sharing of code and data, for novel research even to be considered for publication. These journals are creating incentives for researchers to partake in open science practices by encouraging replication studies for publication. As a result, moving forward we can expect to see stronger research related to bilingual speech production, replications – both successful and failed – of our seminal findings, and larger, more robust studies with increased sample sizes.

18.8 Conclusion

In the present chapter I have provided an overview of the principle aims of speech production research involving adult language learners. The chapter covered in detail important areas of inquiry, such as the nature of age effects on L2 production outcomes, ultimate attainment, and the role of target language input and L1/L2 use in adult bilingual development. I also reviewed some of the fundamental questions that have been asked and are being asked currently, such as how L2 phonetic categories develop over time or how production-perception links develop in bilingualism. Importantly, I also highlighted what I believe to be the principle methodological issues for the field and how we might resolve them moving forward.

References

Aoyama, K., Flege, J. E., Guion, S. G., Akahane-Yamada, R., & Yamada, T. (2004). Perceived phonetic dissimilarity and L2 speech learning: The case of Japanese /r/ and English /l/ and /r/. *Journal of Phonetics*, *32*(2), 233–250.

Berry, W. D. & Feldman, S. (1985). *Multiple Regression in Practice.* Newbury Park, CA: Sage.

Best, C. T. & Tyler, M. D. (2007). Nonnative and second-language speech perception. In O.-S. Bohn & M. J. Munro, eds., *Language Experience in Second Language Speech Learning: In Honor of James Emil Flege.* John Benjamins Publishing Company: Amsterdam, pp. 13–34.

Bley-Vroman, R. (1983). The comparative fallacy in interlanguage studies: The case of systematicity. *Language Learning*, *33*(1), 1–17.

Bongaerts, T., Summeren, C. van, Planken, B., & Schils, E. (1997). Age and ultimate attainment in the pronunciation of a foreign language. *Studies in Second Language Acquisition*, 19(4), 447–465.

Bosch, L. & Ramon-Casas, M. (2011). Variability in vowel production by bilingual speakers: Can input properties hinder the early stabilization of contrastive categories? *Journal of Phonetics*, *39*(4), 514–526.

Brysbaert, M. (2020). Power considerations in bilingualism research: Time to step up our game. *Bilingualism: Language and Cognition*, 24(5), 813–818.

Caramazza, A., Yeni-Komshian, G., Zurif, E. B., & Carbone, E. (1973). The acquisition of a new phonological contrast: The case of stop consonants in French-English bilinguals. *The Journal of the Acoustical Society of America*, *54*(2), 421–428.

Casillas, J. V. (2015). Production and Perception of the /i/-/ɪ/ Vowel Contrast: The Case of L2-Dominant Early Learners of English. *Phonetica*, *72*(2–3), 182–205.

Casillas, J. V. (2019). Semantic processing triggers cross-linguistic interference during early phonetic category development. In S. Calhoun, P. Escudero, M. Tabain, & Paul Warren, eds., *Proceedings of the 19th International Congress of Phonetic Sciences.* Melbourne, Australia, pp. 3348–3352.

Casillas, J. V. (2020a). Phonetic category formation is perceptually driven during the early stages of adult L2 development. *Language and Speech*, ***63***(3), 550–581.

Casillas, J. V. (2020b). The longitudinal development of fine-phonetic detail: Stop production in a domestic immersion program. *Language Learning*, *70*(3), 768–806.

Casillas, J. V. & Simonet, M. (2016). Production and perception of the English /æ/–/ɑ/ contrast in switched-dominance speakers. *Second Language Research*, *32*(2), 171–195.

Chang, C. B. (2012). Rapid and multifaceted effects of second-language learning on first-language speech production. *Journal of Phonetics*, *40*(2), 249–268.

Chang, C. B. (2013). A novelty effect in phonetic drift of the native language. *Journal of Phonetics*, *41*(6), 520–533.

Chen, M. (1970). Vowel length variation as a function of the voicing of the consonant environment. *Phonetica*, *22*(3), 129–159.

Cook, V. (1999). Going beyond the native speaker in language teaching. *TESOL Quarterly*, *33*(2), 185–209.

de Leeuw, E., Stockall, L., Lazaridou-Chatzigoga, D., & Gorba Masip, C. (2021). Illusory vowels in Spanish–English sequential bilinguals: Evidence that accurate L2 perception is neither necessary nor sufficient for accurate L2 production. *Second Language Research*, *37*(4), 587–618.

Diehl, R. L., Lotto, A. J., & Holt, L. L. (2004). Speech perception. *Annual Review of Psychology*, *55*, 149–179.

Díaz-Campos, M. (2004). Context of learning in the acquisition of Spanish second language phonology. *Studies in Second Language Acquisition*, *26*(2), 249–274.

Díaz-Campos, M. (2006). The effect of style in second language phonology: An analysis of segmental acquisition in study abroad and regular-classroom students. In C. A. Klee & T. L. Face, eds., *Selected Proceedings of the 7th Conference on the Acquisition of Spanish and Portuguese As First and Second Languages. Somerville, MA: Cascadilla Press*, pp. 26–39.

Ellis, P. D. (2010). *The Essential Guide to Effect Sizes: Statistical Power, Meta-analysis, and the Interpretation of Research Results.* Cambridge University Press.

Elvin, J., Williams, D., & Escudero, P. (2016). The relationship between perception and production of Brazilian Portuguese vowels in European Spanish monolinguals. *Loquens: Spanish Journal of Speech Sciences*, *3*, 1–10.

Evans, B. G. & Alshangiti, W. (2018). The perception and production of British English vowels and consonants by Arabic learners of English. *Journal of Phonetics*, *68*, 15–31.

Face, T. L. (2018a). Ultimate attainment in Spanish spirantization: The case of U.S.-born immigrants in Spain. *Spanish in Context*, *15*(1), 27–53.

Face, T. L. (2018b). Ultimate attainment of Spanish rhotics by native English-speaking immigrants to Spain. *Language and Migration*, *2*, 57–80.

Face, T. L. (2021). Ultimate attainment of Spanish laterals by native English-speaking immigrants in Spain. *Journal of Second and Multiple Language Acquisition*, *9*, 167–180.

Face, T. L., & Menke, M. (2020). L2 acquisition of Spanish VOT by English-speaking immigrants in Spain. *Studies in Hispanic and Lusophone Linguistics*, *13*(2), 361–389.

Flege, J. E. (1987). The production of "new" and "similar" phones in a foreign language: Evidence for the effect of equivalence classification. *Journal of Phonetics*, *15*(1), 47–65.

Flege, J. E. (1995). Second language speech learning: Theory, findings, and problems. In W. Strange, ed., *Speech Perception and Linguistic Experience Issues in Cross-language Research.* Timonium, MD: York Press, pp. 229–273.

Flege, J. E. (2002). Interactions between the native and second-language phonetic systems. In T. Burmeister, T. Piske, & A. Rohde, eds., *An Integrated View of Language Development: Papers in Honor of Henning Wode.* Trier, Germany: Wissenschaftlicher Verlag, pp. 217–244.

Flege, J. E. (2003). Assessing constraints on second-language segmental production and perception. In N. O. Schiller & A. S. Meyer, eds., *Phonetics and Phonology in Language Comprehension and Production Differences and Similarities*, Berlin: Mouton de Gruyter, pp. 319–355.

Flege, J. E. (2012). The role of input in second language (L2) speech learning. Talk given at the 6*th International Conference on Native and Non-native Accents of English*, Lodz, Poland.

Flege, J. E. & Bohn, O.-S. (2021). The Revised Speech Learning Model (SLM-r). In R. Wayland, ed., *Second Language Speech Learning: Theoretical and Empirical Progress.* Cambridge, MA: Cambridge University Press, pp. 3–83.

Flege, J. E., Bohn, O.-S., & Jang, S. (1997a). Effects of experience on non-native speakers' production and perception of English vowels. *Journal of Phonetics*, *25*(4), 437–470.

Flege, J. E. & Eefting, W. (1987). Production and perception of English stops by native Spanish speakers. *Journal of Phonetics*, *15*, 67–83.

Flege, J. E. & Eefting, W. (1988). Imitation of a VOT continuum by native speakers of English and Spanish: Evidence for phonetic category formation. *The Journal of the Acoustical Society of America*, *83*(2), 729–740.

Flege, J. E., Frieda, E. M., & Nozawa, T. (1997b). Amount of native-language (L1) use affects the pronunciation of an L2. *Journal of Phonetics*, *25*(2), 169–186.

Flege, J. E., Frieda, E. M., Walley, A. C., & Randazza, L. A. (1998). Lexical factors and segmental accuracy in second language speech production. *Studies in Second Language Acquisition*, *20*(2), 155–187.

Flege, J. E. & Hillenbrand, J. (1984). Limits on phonetic accuracy in foreign language speech production. *The Journal of the Acoustical Society of America*, *76*(3), 708–721.

Flege, J. E. & Liu, S. (2001). The effect of experience on adults' acquisition of a second language. *Studies in Second Language Acquisition*, *23*(4), 527–552.

Flege, J. E., MacKay, I. R. A., & Meador, D. (1999). Native Italian speakers' perception and production of English vowels. *The Journal of the Acoustical Society of America*, *106*(5), 2973–2987.

Flege, J. E., Munro, M. J., & MacKay, I. R. A. (1995). Factors affecting strength of perceived foreign accent in a second language. *The Journal of the Acoustical Society of America*, *97*, 3125–3134.

Flege, J. E. & Port, R. (1981). Cross-language phonetic interference: Arabic to English. *Language and Speech*, *24*(2), 125–146.

Flege, J. E. & Wayland, R. (2019). The role of input in native Spanish late learners' production and perception of English phonetic segments. *Journal of Second Language Studies*, *2*(1), 1–33.

Freed, B. F. (1995). *Second Language Acquisition in a Study Abroad Context.* John Benjamins Publishing.

Gelman, A. & Loken, E. (2014). The statistical crisis in science: Data-dependent analysis–a" garden of forking paths"–explains why many statistically significant comparisons don't hold up. *American Scientist*, *102*(6), 460–466.

González-Bueno, M. (1997). Voice onset time in the perception of foreign accent by native listeners of Spanish. *International Review of Applied Linguistics in Language Teaching*, *35*(4), 251–262.

Gorba, C. & Cebrian, J. (2021). The role of L2 experience in L1 and L2 perception and production of voiceless stops by English learners of Spanish. *Journal of Phonetics*, *88*, 1–25.

Gustafson, E., Engstler, C., & Goldrick, M. (2013). Phonetic processing of non-native speech in semantic vs non-semantic tasks. *The Journal of the Acoustical Society of America*, *134*(6), EL506–EL512.

Herd, W. J., Walden, R. L., Knight, W. L., & Alexander, S. N. (2015). Phonetic drift in a first language dominant environment. *The Journal of the Acoustical Society of America*, *137*(4), 2384–2384.

Holliday, J. J. (2015). A longitudinal study of the second language acquisition of a three-way stop contrast. *Journal of Phonetics*, *50*, 1–14.

Huffman, M. K. & Schuhmann, K. (2015a). Effect of early L2 learning on L1 stop voicing. *Proceedings of Meetings on Acoustics*, *23*(1), 060007.

Huffman, M. K. & Schuhmann, K. (2015b). Individual differences in phonetic drift by English-speaking learners of Spanish. *The Journal of the Acoustical Society of America*, *137*(4), 2268–2268.

Kartushina, N. & Frauenfelder, U. H. (2014). On the effects of L2 perception and of individual differences in L1 production on L2 pronunciation. *Frontiers in Psychology*, *5*.

Kingston, J. & Diehl, R. L. (1994). Phonetic knowledge. *Language*, *70*, 419–454.

Lang, B. & Davidson, L. (2019). Effects of exposure and vowel space distribution on phonetic drift: Evidence from American English learners of French. *Language and Speech*, *62*(1), 30–60.

Lenneberg, E. H. (1967a). *Biological Foundations of Language.* New York: Wiley.

Lenneberg, E. H. (1967b). The biological foundations of language. *Hospital Practice*, *2*(12), 59–67.

Levis, J. M. (2005). Changing contexts and shifting paradigms in pronunciation teaching. *TESOL Quarterly*, *39*(3), 369–377.

Lewis-Beck, M. (1980). Multiple regression. In M. Lewis-Beck, ed., *Applied regression: An Introduction.* Newbury Park, CA: Sage, pp. 47–74.

Lisker, L. & Abramson, A. S. (1964). A Cross-language Study of Voicing in Initial Stops: Acoustical Measurements. *Word*, ***20***(3), 384–422.

Mennen, I. (2015). Beyond segments: Towards a L2 intonation learning theory. In E. Delais-Roussarie, S. Herment, & M. Avanzi, eds., *Prosody and Language in Contact.* Springer, pp. 171–188.

Mitchell, R., Myles, F., & Marsden, E. (2019). *Second Language Learning Theories.* Routledge.

Mora, J. C. & Nadeu, M. (2012). L2 effects on the perception and production of a native vowel contrast in early bilinguals. *International Journal of Bilingualism*, *16*(4), 484–500.

Nagle, C. L. (2017a). A Longitudinal Study of Voice Onset Time Development in L2 Spanish Stops. *Applied Linguistics*, *54*(5), 13–23.

Nagle, C. L. (2017b). Individual developmental trajectories in the L2 acquisition of Spanish spirantization. *Journal of Second Language Pronunciation*, *3*(2), 218–241.

Nagle, C. L. (2021). Assessing the state of the art in longitudinal L2 pronunciation research: Trends and future directions. *Journal of Second Language Pronunciation*, *7*(2), 154–182.

Nagle, C. L., & Baese-Berk, M. M. (2021). Advancing the state of the art in L2 speech perception-production research: Revisiting theoretical assumptions and methodological practices. *Studies in Second Language Acquisition*, 4(2), 580–605.

Nozari, N., Kittredge, A. K., Dell, G. S., & Schwartz, M. F. (2010). Naming and repetition in aphasia: Steps, routes, and frequency effects. *Journal of Memory and Language*, *63*(4), 541–559.

Ortega, L. (2013). Ways forward for a bi/multilingual turn in SLA. In S. May, ed., *The Multilingual Turn: Implications for SLA, TESOL, and Bilingual Education.* New York: Routledge, pp. 32–53.

Oyama, S. (1976). A sensitive period for the acquisition of a nonnative phonological system. *Journal of Psycholinguistic Research*, *5*, 261–283.

Peperkamp, S. & Bouchon, C. (2011). The relation between perception and production in L2 phonological processing. In P. Cosi, R. De Mori, G. Di Fabbrizio, & R. Pieraccini, eds., *Proceedings of Interspeech 2011).* Rundle Mall: Causal Productions, pp. 161–164.

Piske, T., Flege, J. E., MacKay, I. R. A., & Meador, D. (2002). The production of English vowels by fluent early and late Italian-English bilinguals. *Phonetica*, *59*(1), 49–71.

Plonsky, L. & Oswald, F. L. (2014). How big is “big”? Interpreting effect sizes in L2 research. *Language Learning*, *64*(4), 878–912.

Port, R., & Mitleb, F. M. (1983). Segmental features and implementation in acquisition of English by Arabic speakers. *Journal of Phonetics*, *11*(3), 219–229.

Rallo Fabra, L. & Romero, J. (2012). Native Catalan learners’ perception and production of English vowels. *Journal of Phonetics*, *40*(3), 491–508.

Raphael, L. J. (2005). Acoustic cues to the perception of segmental phonemes. In D. Pisoni & R. Remez, eds., *The Handbook of Speech Perception.* Oxford: Wiley-Blackwell, pp. 182–206.

Rotello, C. M., Heit, E., & Dubé, C. (2015). When more data steer us wrong: Replications with the wrong dependent measure perpetuate erroneous conclusions. *Psychonomic Bulletin & Review*, *22*(4), 944–954.

Sakai, M. (2018). Moving towards a bilingual baseline in second language phonetic research. *Journal of Second Language Pronunciation*, *4*(1), 11–45.

Saville-Troike, M. (2005). *Introducing Second Language Acquisition.* Cambridge University Press.

Scovel, T. (1969). Foreign accents, language acquisition, and cerebral dominance. *Language Learning*, *19*(3–4), 245–253.

Scovel, T. (1988). *A Time to Speak: A Psycholinguistic Inquiry into the Critical Period for Human Speech.* Rowley, MA: Newbury House.

Seliger, H. (1978). Implications of a multiple critical periods hypothesis for second language learning. In W. Ritchie, ed., *Second Language Acquisition Research: Issues and Implications.* New York: Academic Press, pp. 11–19.

Simmons, J. P., Nelson, L. D., & Simonsohn, U. (2011). False-positive psychology: Undisclosed flexibility in data collection and analysis allows presenting anything as significant. *Psychological Science*, *22*(11), 1359–1366.

Starns, J. J., Cataldo, A. M., Rotello, C. M. et al. (2019). Assessing theoretical conclusions with blinded inference to investigate a potential inference crisis. *Advances in Methods and Practices in Psychological Science*, *2*(4), 335–349.

Stevens, J. (2001). Study abroad learners’ acquisition of the Spanish voiceless stops. *MIFLC Review*, *10*, 137–151.

Stölten, K., Abrahamsson, N., & Hyltenstam, K. (2015). Effects of age and speaking rate on voice onset time: The production of voiceless stops by near-native L2 speakers. *Studies in Second Language Acquisition*, *37*(1), 71–100.

Sundara, M., Polka, L., & Baum, S. (2006). Production of coronal stops by simultaneous bilingual adults. *Bilingualism: Language and Cognition*, *9*(01), 97–114.

Trofimovich, P. & Baker, W. (2006). Learning second language suprasegmentals: Effect of L2 experience on prosody and fluency characteristics of L2 speech. *Studies in Second Language Acquisition*, *28*(1), 1–30.

Trofimovich, P. & Baker, W. (2007). Learning prosody and fluency characteristics of second language speech: The effect of experience on child learners' acquisition of five suprasegmentals. *Applied Psycholinguistics*, *28*(2), 251–276.

Trubetzkoy, N. S. (1969). *Principles of phonology*. Berkeley: University of California Press.

Van Lancker Sidtis, D., Cameron, K., & Sidtis, J. J. (2012). Dramatic effects of speech task on motor and linguistic planning in severely dysfluent parkinsonian speech. *Clinical Linguistics & Phonetics*, *26*(8), 695–711.

Van Leussen, J.-W. & Escudero, P. (2015). Learning to perceive and recognize a second language: The L2LP model revised. *Frontiers in Psychology*, *6*, 1–12.

Walsh, T. & Diller, K. (1981). Neurolinguistic considerations on the optimum age for second language learning. In K. Diller, ed., *Individual Differences and Universals in Language Learning Aptitude.* Cambridge, MA: MIT Press, pp. 3–21.

Weinreich, U. (1953). *Languages in Contact: Findings and Problems.* The Hague: Mouton/New York, NY: Publications of the Linguistic Circle of New York.